

RESEARCH ARTICLE OPEN ACCESS

Universal Behavior Screening and Early Warning System Indicators in Middle Schools

Emily Graybill¹  | Scott Lewis² | Ella Anghel¹  | Sofia Awan² | Brian Barger² | Ashley Salmon²

¹Department of School Psychology, Rutgers University, New Brunswick, New Jersey, USA | ²School of Public Health, Georgia State University, Atlanta, Georgia, USA

Correspondence: Emily Graybill (emily.graybill@rutgers.edu)

Received: 1 July 2024 | **Revised:** 21 February 2025 | **Accepted:** 23 March 2025

Keywords: early warning systems | mental health | middle school | universal screening

ABSTRACT

Early warning indicator and intervention systems (EWS) have been promoted to identify students at risk of school underperformance or dropout. Current EWS systems typically include administrative data on attendance, behavior incidents requiring disciplinary action, and course performance. This study tested whether specific emotional and behavioral risk symptoms measured by a student self-report universal screener administered in the fall can predict the three EWS indicators after controlling for fall behavioral incidents and whether they account for some of the variance in EWS attributed to demographic characteristics. Using data from 3307 middle school students, we found that after accounting for fall disciplinary issues, conduct problems predicted poorer student outcomes, but hyperactivity/inattention was predictive of better course performance. Peer problems predicted lower performance in some courses, while emotional problems predicted better performance as well as fewer behavioral issues. We also found that these symptoms accounted for some of the variance in EWS attributed to race and gender. The results suggest that student self-report universal screening can complement existing EWS measures and potentially identify at-risk students who may not otherwise be identified through traditional EWS indicators.

1 | Introduction

School dropout is a public health issue with community- and individual-level risk and protective factors (Lansford et al. 2016; Shin et al. 2012). Over 2 million high school students in the United States (U.S.) drop out of school each year. This equates to over 1000 students per school day (National Center for Education Statistics [NCES] 2023). These data underscore the urgent need for schools to build large data systems within a public health framework that facilitate dropout prevention activities and early identification and intervention for students at risk of dropping out (Freudenberg and Ruglis 2007).

Increasingly, schools use their vast sets of administrative data to identify students at risk for dropout and implement early

intervention to mitigate that risk (Neild et al. 2007; Regional Educational Laboratory Pacific 2016). Early Warning Indicator and Intervention Systems (EWS) are data systems used to identify students at risk of dropout and low school engagement (Balfanz and Byrnes 2019). They grew from the idea that students show early signs of disengagement with school and that early intervention by educators may prevent dropout (Bruce et al. 2011; Neild et al. 2007). In a sign of their effectiveness, the U.S. Department of Education has recognized the use of EWSs to identify students at an increased risk of dropping out of school (U.S. Department of Education 2016).

Most commonly, EWSs use schools' administrative data, specifically attendance, behavioral issues, and course performance,

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2025 The Author(s). *Psychology in the Schools* published by Wiley Periodicals LLC.

Summary

- School district early warning system indicators often do not include a measure of emotional and behavior risk or a student self-report measure of early risk.
- Among middle schoolers, students with internalizing concerns such as peer and emotional problems may not be at risk on other early warning system indicators – such as attendance, behavior, and course performance – and would likely be missed as needing early intervention without behavior screening data.
- Student self-report universal behavior screening measuring internalizing and externalizing concerns can complement existing early warning systems to identify at-risk students early, particularly those with internalizing concerns who may not be at risk on other early warning system indicators.

also known as the “ABCs”, to identify at-risk students (Bruce et al. 2011). These three ABC domains are widely used because they are highly predictive of high school completion (Neild et al. 2007; Pinkus 2008; Wentworth and Nagaoka 2020). However, relying solely on ABCs to identify at-risk students has several disadvantages. First, the data are only available several months into the school year at the earliest, which delays the start of interventions (de Vasconcelos et al. 2023) and promotes a “wait-to-fail” model, flagging students who are already struggling (Cavanaugh 2016). While ABC indicators from the prior spring can be used to inform interventions in the following year, information from the previous year might not reflect a student’s current difficulties, and it still allows schools to intervene only among students who have struggled for long periods of time.

Second, traditional EWS indicators do not include student self-report data. Most middle and high school students can recognize their own disengagement (Archambault et al. 2009) even if it is not observable by teachers or administrative data (Lovelace et al. 2017). Indeed, some research has explored the relationship between student self-report screening data and student outcomes, suggesting that student self-report data add unique information to the identification of student risk (Margherio et al. 2019).

Finally, consistent findings point to the association between mental health and dropout risk (Hjorth et al. 2016). Cross-sectional and longitudinal research shows that untreated mental health concerns are one risk factor for leaving school (Butterworth and Leach 2018; Hjorth et al. 2016; Lawrence and Adebawale 2023). Conversely, both experimental and associational studies show that students are less likely to drop out if they receive needed mental health support and have positive mental wellness (Heller et al. 2013; Lawrence and Adebawale 2023), and students who had recovered from earlier mental health concerns are at a lower risk of dropping out. These findings suggest that by intervening early with youth with such concerns, educators may be able to prevent school dropout (Rodriguez and Conchas 2009), highlighting the importance of early identification of and intervention for students’ mental health concerns. Yet, the

typical EWS in schools only includes observed behavioral problems like discipline referrals instead of emotional risk indicators such as universal behavioral and emotional risk screening (hereafter referred to as universal behavior screening). There is also minimal empirical literature on the inclusion of an emotional and behavior risk data set within districts’ EWS to guide schools on using such data to inform prevention and early intervention activities (Davis et al. 2019).

1.1 | Universal Behavior Screening and Early Warning Systems

Universal behavior screeners can address many of the issues that exist with traditional EWSs. Screeners can be easily administered early in the school year, allowing for earlier identification and intervention. They reveal students’ perspectives on their own emotional and behavioral status (when using a student self-report screener), complementing the external perspective of administrative data. Moreover, they allow for the detection of students with internalizing concerns who are usually under-identified using traditional EWSs despite their heightened risk (e.g., Graybill et al. 2022). School-based screeners often consist of externalizing and internalizing subscales, and some screeners further expand on these general subscales by including subscales or items targeting specific symptoms such as peer rejection, hyperactivity, and academic behavior (e.g., Drummond 1994; Izumi and Eklund 2023).

In recent years, more research has examined how universal screeners relate to EWS indicators. These studies have shown that emotional and behavioral risk screeners can predict office discipline referrals (ODRs; Gregory et al. 2020; Graybill et al. 2022), attendance (Jones et al. 2020), and academic attainment (Eklund et al. 2009). However, existing studies have rarely examined all three ABC indicators together alongside student self-report screening data. Since currently, EWS are used together in schools to identify at-risk students, it is important to also test whether screeners can predict all three of them together.

In addition, these studies usually focus on internalizing and externalizing scores, even though many screeners offer subscales targeting more specific issues. Investigating how specific subscales relate to the ABC data can help practitioners understand the complex relationship between risk indicators and focus on the most informative subscales data to include in their EWSs. For example, Young et al.’s (2021) work examined specific screener scores (aggression, peer rejection, flat emotion, etc.) and their association with attendance, academic performance, and in-school and out-of-school suspensions. They found that even within the externalizing and internalizing groups of symptoms, there were specific subscales whose relationships with ABCs diverged in direction and strength. However, Young et al. (2021) only used a teacher-report screener. Exploring this effect with a student-report screener is warranted, as it may offer unique benefits with less burden on teachers.

1.2 | Use of Office Discipline Referrals as Screeners

Schools typically use their ODR data as their behavior metric in their EWS. Some studies show that ODRs accrued by a student in early fall predict spring ODRs and other end-of-year outcomes (McIntosh et al. 2010). However, the use of Fall ODR data still means that interventions can be applied in November at the earliest, missing out on months of potential preventative work with at-risk students. Therefore, it is important to test the advantage of universal screening relative to Fall ODR count in predicting student outcomes.

TABLE 1 | Sample's demographics.

		Frequency	Percentage
Gender	Female	1437	47
	Male	1511	50
	Nonbinary	30	1
	Prefer not to say	57	2
	Unknown	2	0.1
Race	American Indian or Alaska Native	45	2
	Asian	25	1
	Black or African American	2340	77
	Hispanic	117	4
	Multi-racial	182	6
	Native Hawaiian or Other Pacific Islander	9	0.3
	Other/Unknown	21	1
	White	298	10

TABLE 2 | Student's performance.

	% Beginning (1)	% Developing (2)	% Proficient (3)	% Distinguished (4)
ELA Achievement	37.91	38.33	21.00	2.76
Math Achievement	50.97	34.84	11.23	2.96

TABLE 3 | Descriptive statistics for the SDQ subscales.

	Mean	Standard deviation	Minimum	Maximum	Range	Skewness	Kurtosis
Emotional problems	3.59	2.49	0	10	10	0.40	−0.63
Conduct problems	2.28	1.87	0	10	10	0.77	0.11
Hyperactivity-inattention	4.07	2.30	0	10	10	0.23	−0.46
Peer problems	3.02	1.85	0	10	10	0.53	0.06

1.3 | Disparities in Dropout Rates and EWS

Dropout rates vary by race, ethnicity, and gender, with American Indian/Alaska Native, Hispanic, and Black youth experiencing the highest rates of dropout (National Center for Educational Statistics [NCES] 2017). Across most races, males are more likely to drop out than females (National Center for Educational Statistics 2017). The same differences in racial and gender data exist with the ABC indicators in EWS. Black students and males are more likely to receive discipline referrals (Liu et al. 2024) and have lower achievement scores (Assari et al. 2021). These disparities and the underlying racial bias justify the need to understand how universal screening and ABC data trends vary by race and gender (Chin et al. 2020). Universal screening data are collected systematically and used to inform a preventative and proactive data-based decision-making process for all students. Data from a universal screening process may not include the inconsistencies and bias inherent in the assignment of ODRs as a discipline strategy or in the review of ODRs to make decisions about individual students (Tara C. Raines et al. 2012; Barger et al. 2020).

1.4 | Current Study

In the current study, we explore the contribution of a student self-report universal behavior screener within an EWS to better understand the value added by including early mental health data in an EWS. More specifically, we examine the relationship between universal behavior screening subscales with the original EWS indicators, attendance, behavior, and course performance in a middle school setting. We also extend existing research by comparing the screeners' subscales to Fall ODRs and as predictors of end-of-year ABCs and by testing whether the universal screening accounts for some of the gender and race disparities in ABCs.

To do so, we analyzed data from students in six middle schools who completed the self-report version of the Strengths and Difficulties Questionnaire (SDQ; Goodman 2001), focusing on the SDQ subscales of emotional problems, peer problems, conduct problems, and hyperactivity-inattention. While many existing studies focus on high school students' data, research on EWS suggests that risk indicators of middle schoolers can predict later dropout risk (Balfanz et al. 2007). As we were mostly

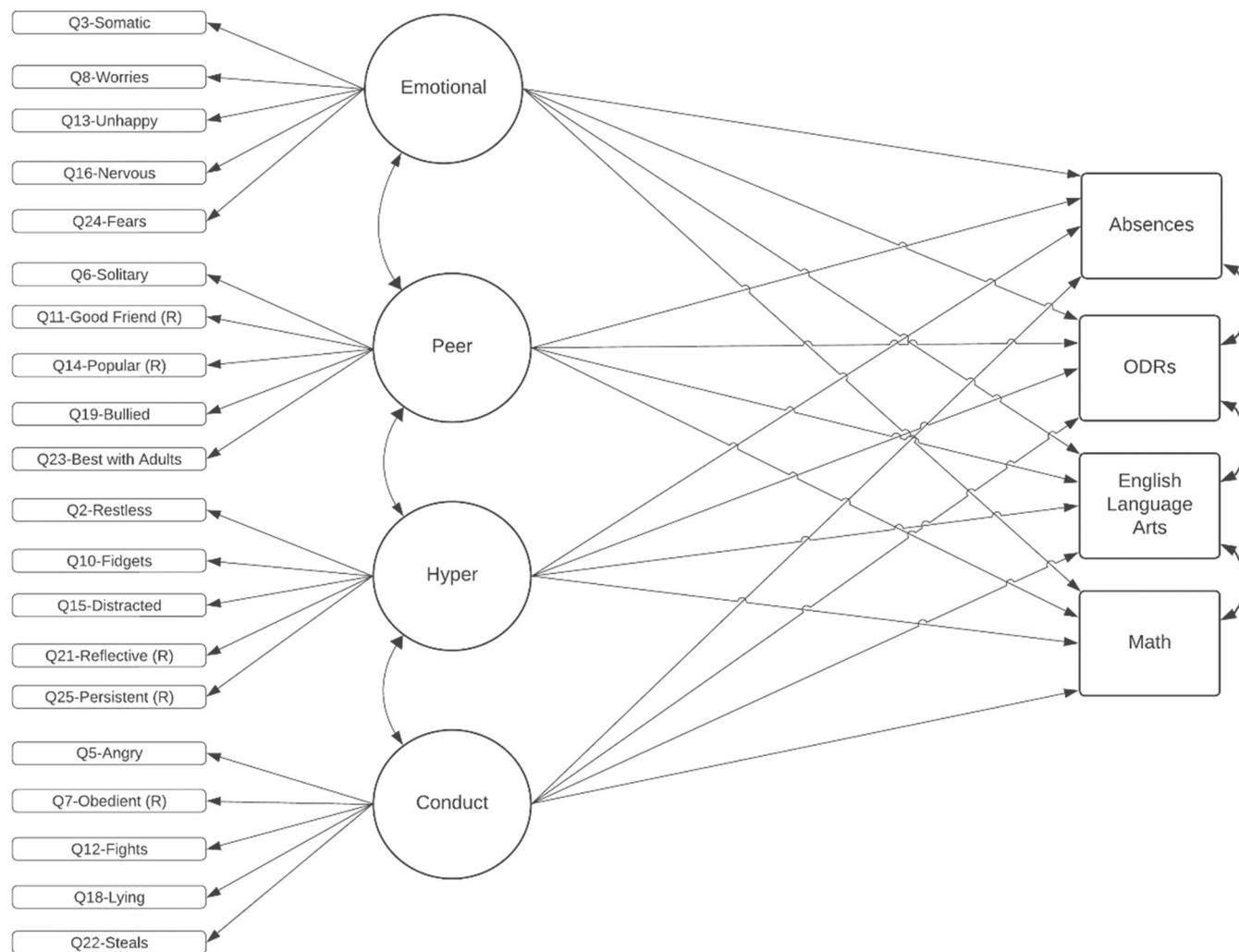


FIGURE 1 | Predicting EWS outcomes from the four SDQ subscales (Model 1). (R) indicates a reverse-coded item.

interested in early risk detection, we focused on middle schoolers. This study may inform future efforts to enrich EWSs to support data-based decision-making within districts' public health tiered system of support framework to facilitate the early identification of and intervention for students at risk of dropout.

Our research questions are:

RQ1 *Do the universal screener subscales predict attendance, behavior, and course performance?*

RQ2 *Do the universal screener subscales predict attendance, behavior, and course performance over and above Fall ODRs?*

RQ3 *Do the universal screeners capture some variance in EWS indicators attributed to gender or race?*

2 | Methods

2.1 | Context

Data for this study were collected as part of a school-based mental health project in the Southeastern U.S. This is a 5-year project

designed to help schools build infrastructure related to mental health, including strengthening the mechanisms of identifying at-risk students and referring them to further support. The schools participating in this project administered the SDQ in September 2021, at the beginning of the 2021–2022 school year, the first year of the project. The screening data were then utilized to inform school-level decision-making within a multi-tiered system of support framework. The schools reviewed school-level trends in the screening data to inform their implementation of schoolwide programs. Individual-level student risk was also reviewed to inform decisions about targeted and individualized interventions. Because these data were collected in Year 1 of a multi-year infrastructure-building project, it is not expected that the implementation of the project influenced trends in the student data reported in this article. Students' guardians provided consent for them to participate in the universal behavior screening process and this study was approved by Rutgers University Institutional Review Board.

In this study, we included data from 3037 students within six middle schools (grades 6th through 8th) who had screening and administrative data available. Approximately 47% of respondents were female, 50% were male, and 3% were either Nonbinary or did not report their gender. Most students were Black or African

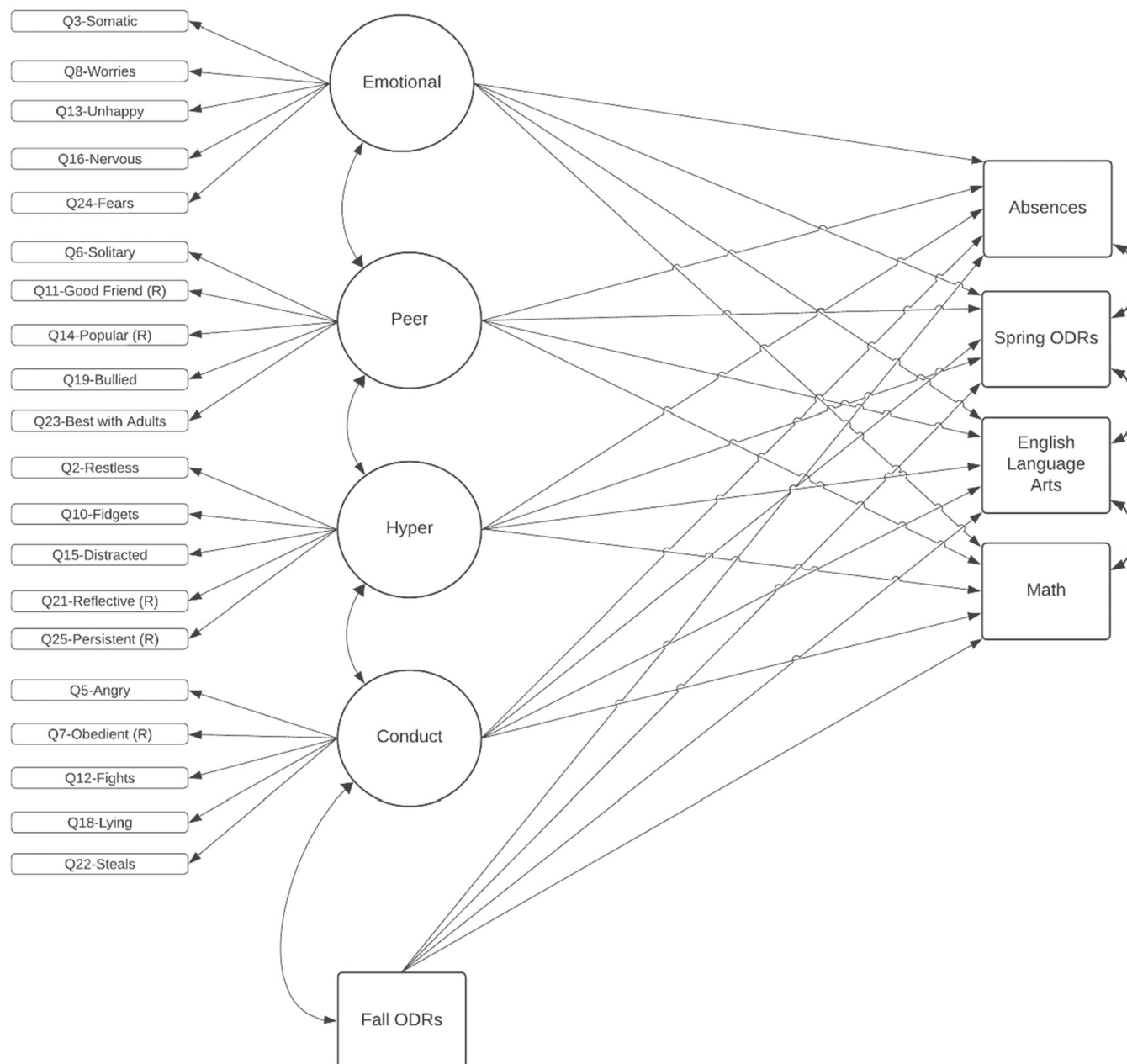


FIGURE 2 | Predicting EWS outcomes from the four SDQ subscales and Fall ODRs (Model 2). (R) indicates a reverse-coded item.

American (76%), representing the demographics of the school district. The remaining 24% of students were grouped as “Other Race.” Black students were used as the reference group in the models exploring differences by race due to Black students comprising the majority of the sample and also being of high interest to the target school district (Johfre and Freese 2021). All students in the sample were eligible for free- or reduced-price lunch. The students’ demographics are also presented in Table 1.

2.2 | Measures

2.2.1 | EWS Data

Data on the students’ attendance, behavior, and course performance were included in the analyses. Attendance data

consisted of both excused and unexcused absences. As research suggests that more than 20 absences a year significantly reduces graduation rates (Bruce et al. 2011), we chose to dichotomize this variable such that students with 20 or more absences were coded as one and all others were coded as 0. Before dichotomization, each student had 14.40 absences on average (standard deviation = 13.36). After dichotomization, 6.99% of students had more than 20 absences. Behavior outcome measures consisted of student-level ODR counts, and they were also dichotomized such that more than two ODRs were coded as 1 and all others were coded as 0, following Bruce et al.’s (2011) recommendations based on a summary of the literature and several case studies. The mean number of ODRs before October was 0.48 (standard deviation = 0.86), with 3.82% of students had at least two ODRs before October. Throughout the year, the mean number of ODRs was 2.53 (standard deviation = 2.09), and

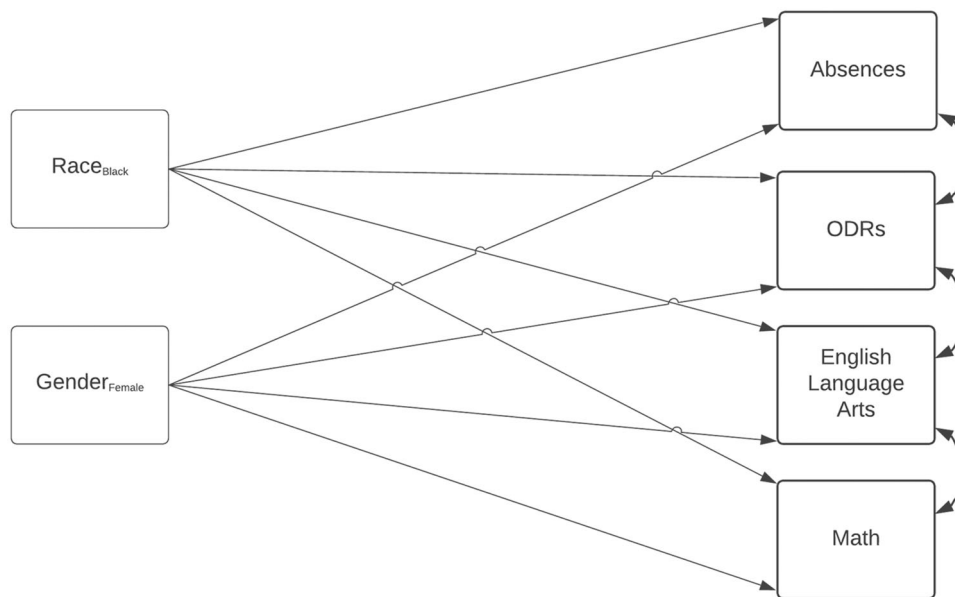


FIGURE 3 | Predicting EWS outcomes from race and gender (Model 3).

17.50% had at least two ODRs. Course performance data included the state-wide end-of-grade test for English language arts (ELA) and math, where performance was categorized by the state-testing system into four categories, representing beginning (1), developing (2), proficient (3), and distinguished (4) learners. Table 2 presents the percentages of the achievement levels.

2.2.2 | Strengths and Difficulties Questionnaire (SDQ)

The SDQ (Goodman 1997) is commonly used to screen for the behavioral and emotional risk of children aged 2–17. It was selected as the screener in this study due to its strong psychometric properties (Goodman 2001) and its affordability for the school district. The SDQ consists of 25 questions answered on a 3-point Likert scale (*Not True*, *Somewhat True*, *Certainly True*). These 25 attributes then inform a five-subscale structure: emotional problems (e.g. ‘I worry a lot’), conduct problems (‘I usually do as I am told,’ reverse coded), hyperactivity-inattention (‘I am restless, I cannot stay still for long’), peer problems (‘I would rather be alone than with people of my age’), and prosocial behavior (‘I am kind to younger children’). All scales were coded such that higher scores represent higher levels (i.e., more mental health concerns). Table 3 presents descriptive statistics of the four scale scores.

2.3 | Data Analysis

Structural equation modeling (SEM) was used to construct our models. The outcomes of these models were absences (dichotomized), ODRs (dichotomized), ELA performance categories, and math performance categories. In the first model (Figure 1), we included twenty manifest variables (items), each loaded on one of four latent variables representing the SDQ subscales (hyperactivity/inattention, conduct problems, emotional problems, and peer problems), five items for each subscale. This model was used to answer RQ1.

The second model (Figure 2) was similar, except we added the dichotomized Fall ODRs as a predictor (i.e., all ODRs before October 31st), and the behavior outcomes included all other ODRs. When using ODRs as a predictor and outcome, we used the same categories reported above. This model was used to answer RQ2.

Finally, to understand trends by demographics, we compared Model 3 (Figure 3), which included only race and gender as predictors of EWS outcomes, to Model 4 (Figure 4), which also includes the SDQ items and subscales. Specifically, we examined the standardized coefficients of race and gender between Model 3 and Model 4 to see if the SDQ subscales account for some of their effect on the EWS outcomes.

All SEM analyses were performed using the lavaan package (version 0.6–17) in R version 4.3.2 (Rosseel 2012). The models were estimated using diagonally weighted least squares, with mean and variance adjustment (WLSMV) to accommodate categorical variables (Graybill et al. 2020; Barger et al. 2020; Li 2015). Following Kline (2015), we estimated Models 1, 2, and 4 iteratively revising them by allowing items to covary until an acceptable fit was obtained. Model 3 only included demographics and no items, so this step was not necessary. We evaluated the models’ fit to the data using comparative fit index (CFI), root mean square error of approximation (RMSEA), and standardized root mean square residual (SRMR). A model was considered as having an acceptable fit if its CFI > 0.90, RMSEA < 0.05, and SRMR < 0.08 (Byrne 1994). We also report the models’ χ^2 , though as this parameter is sensitive to sample size, we did not use it for fit evaluation (Kline 2015). Note that Model 3 was just identified, so meaningful fit statistics were not reported for that model.

3 | Results

A zero-order correlation table is presented in Appendix A. All models were well-fitting after minor modifications. In all

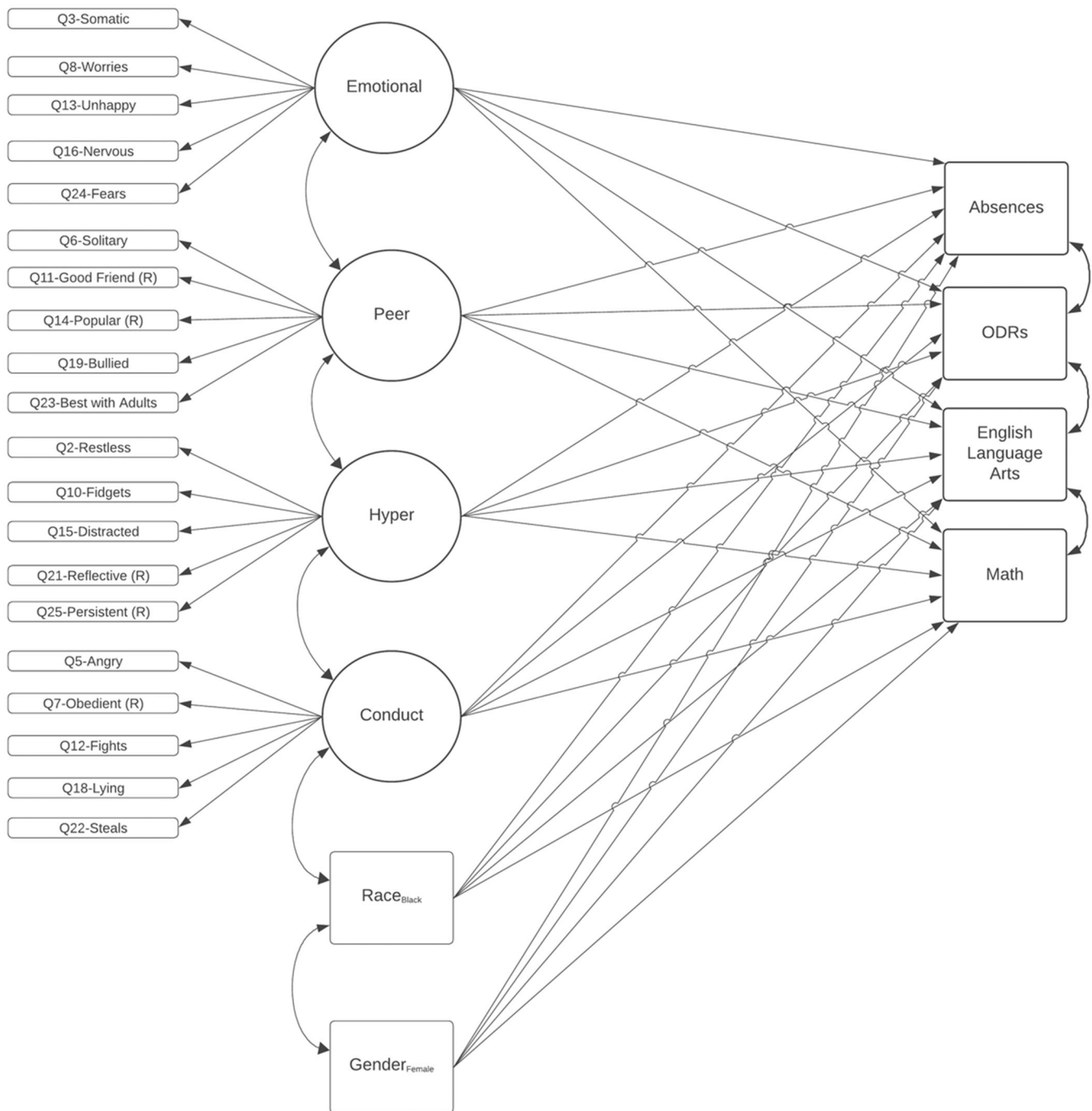


FIGURE 4 | Predicting EWS outcomes from race, gender, and the four SDQ subscales (Model 4). (R) indicates a reverse-coded item.

models, we allowed two pairs of items on the peer problems subscale to covary (solitary-bullied, good friend-popular), and for three items in the conduct problems (obedient) and the hyperactivity-inattention (reflective, persistent) subscales to covary. In Model 4, we also had to allow all predictors to covary (i.e., race, gender, and all subscales).

Model 1 was used to answer RQ1 by checking whether the screener subscales predict our ABC indicators. The final fit statistics for this model were $\chi^2(223) = 1839.82$, $p < 0.001$, CFI = 0.946, RMSEA = 0.049, SRMR = 0.06. Figure 5 presents the standardized coefficients for the significant paths in Models 1; Note that all of the figures presented in this paper do not

show the covarying items for a cleaner presentation. The other coefficients, along with specific p -values and other statistics, are available in Appendix B.

We found that emotional problems predicted lower rates of ODRs and higher ELA achievement ($\beta = -0.422$, 0.170 , respectively). Peer problems only predicted lower ELA scores ($\beta = -0.145$). Hyperactivity-Inattentiveness predicted higher achievement both in ELA ($\beta = 0.313$) and in math ($\beta = 0.328$). The conduct problems subscale was the strongest predictor of all four outcomes: it was positively associated with absences ($\beta = 0.172$) and ODRs ($\beta = 0.534$) and negatively associated with ELA ($\beta = -0.457$) and math scores ($\beta = -0.438$).

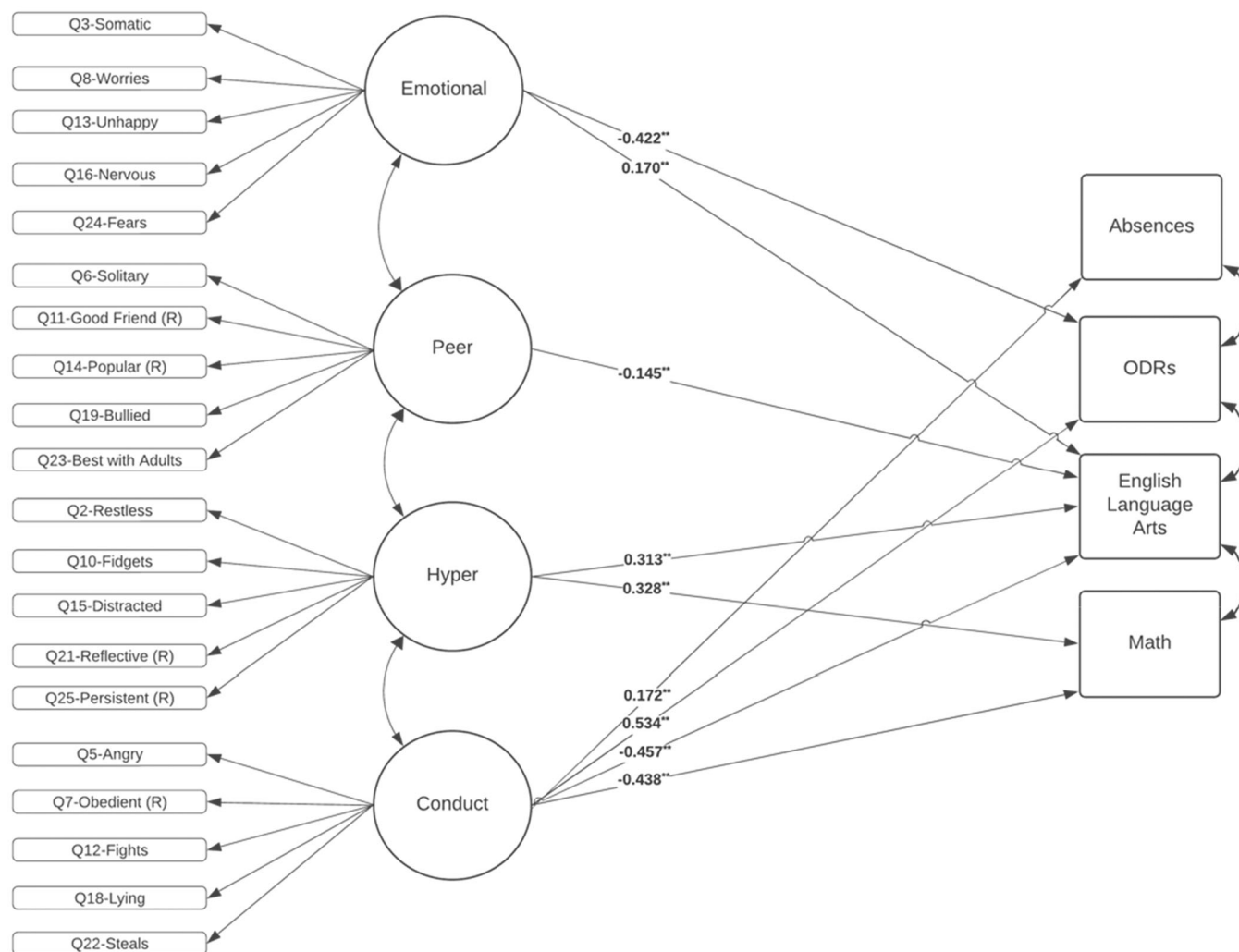


FIGURE 5 | Model 1 predicting attendance, behaviors, and ELA and math performance from the four SDQ subscales. (R) indicates a reverse-coded item.

Model 2 was structured to test whether the relationships presented in Model 1 exist after controlling for Fall ODRs, thus answering RQ2. The fit statistics of Model 2 were $\chi^2(241) = 1910.49$, $p < 0.001$, CFI = 0.945, RMSEA = 0.048, SRMR = 0.06. It is presented in Figure 6 (Appendix C contains a full description of the model). We found that in Model 2, Fall ODRs were also a significant predictor of all outcomes: absences ($\beta = 0.337$), Spring ODRs ($\beta = 0.421$), ELA scores ($\beta = -0.243$), and math scores ($\beta = -0.243$). The screener subscales generally remained significant predictors in the same direction as in Model 1, though their effect size was slightly weaker. There were, however, several exceptions. The peer problems subscale now significantly predicted absences ($\beta = 0.124$) and Spring ODRs ($\beta = 0.118$), and was a stronger predictor of ELA scores ($\beta = -0.172$). In addition, the conduct problem subscale no longer significantly predicted absences but was still a strong predictor of ODRs and achievement scores. When comparing the subscales' effect size to that of the fall ODRs, we found that fall ODRs had the strongest effects on absences and spring ODRs, but that the conduct problems and hyperactivity subscales were better predictors of both achievement variables. The other screener subscales had weaker effects.

Given the surprising results with respect to the hyperactivity-inattentiveness subscale, we suspected a potential suppression effect, where the association between hyperactivity-inattentiveness and the outcomes are accounted for by the conduct problems scores. First, we examined the bivariate correlations between the hyperactivity-inattentiveness scores and the outcomes; we found significant and positive, albeit weak, correlations with absences ($r = 0.06$) and the number of ODRs ($r = 0.05$), and nonsignificant correlations with ELA and math scores. Running models 1 and 2 without conduct problems also showed that hyperactivity-inattentiveness significantly predicted more absences ($\beta = 0.089$ in Model 1; n.s. in Model 2) and ODRs ($\beta = 0.241$, 0.106 for Models 1 and 2, respectively). However, hyperactivity-inattentiveness was still positively associated with ELA ($\beta = 0.068$, 0.145) and Math achievement ($\beta = 0.091$, 0.167).

Finally, we compared Model 3 containing only race and gender as predictors to Model 4 adding the screener subscales to see whether they account for some of the observed relationships between demographics and the EWS outcomes, thus answering RQ3. Models 3 and 4 are presented in Figures 7 and 8, respectively (full results are available in Appendix D). The fit

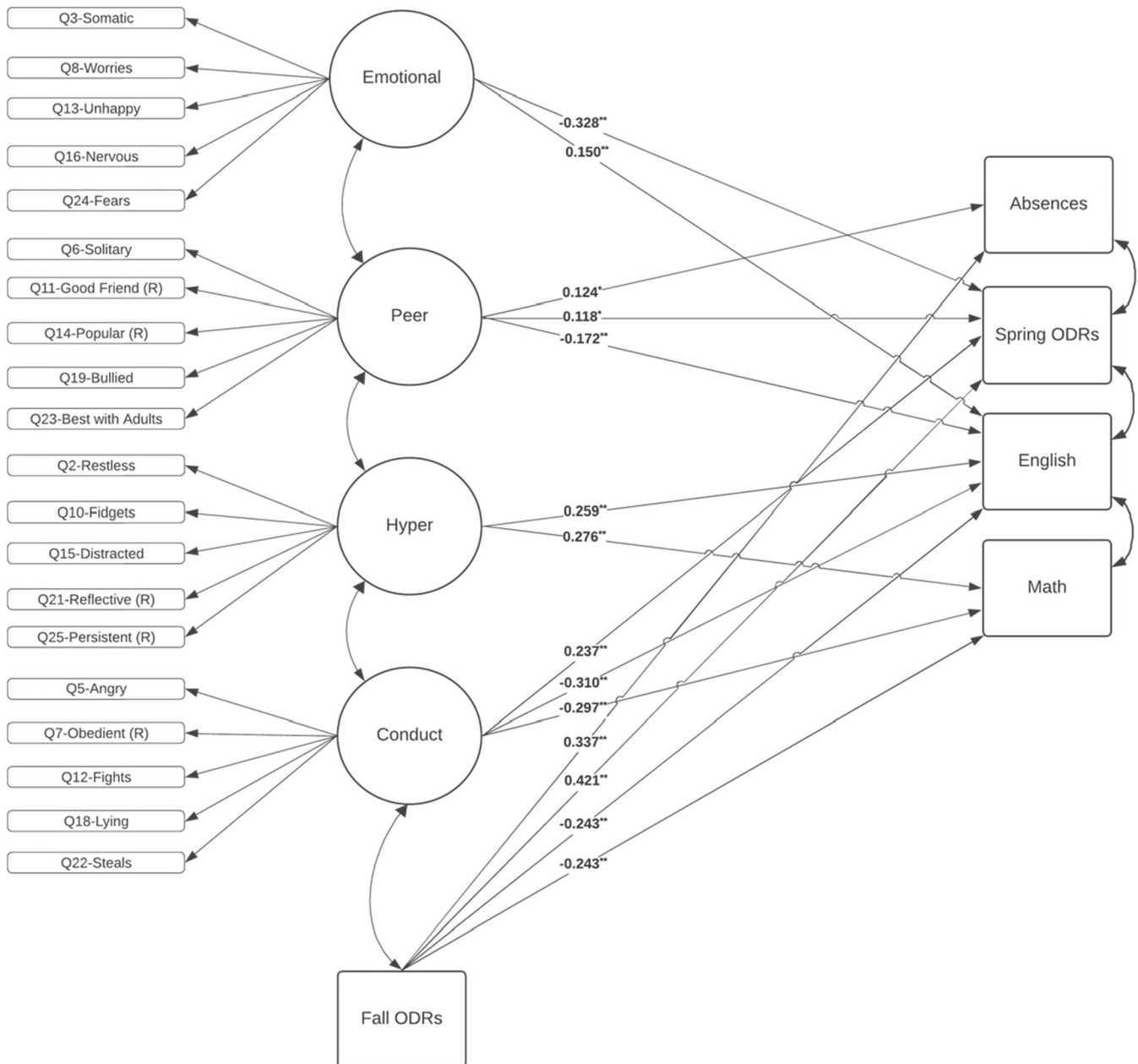


FIGURE 6 | Model 2 predicting attendance, behaviors, and ELA and math performance from the four SDQ subscales and Fall ODRs. (R) indicates a reverse-coded item.

statistics for Model 4 were $\chi^2(256) = 2038.20$, $p < 0.001$, CFI = 0.943, RMSEA = 0.048, SRMR = 0.06.

In Model 3, race was a significant predictor of Spring ODRs ($\beta = 0.203$), ELA scores ($\beta = -0.223$), and math scores ($\beta = -0.223$), but not absences. That is, Black students had more ODRs and lower achievement scores than other students. Gender was a significant predictor of Spring ODRs ($\beta = -0.144$) and ELA scores ($\beta = 0.096$), but not absences or math scores. That is, females had fewer ODRs and higher ELA scores than males.

In Model 4, after accounting for the screener subscales, Black students still had significantly more Spring ODRs ($\beta = 0.171$) and lower ELA and math scores ($\beta = -0.193$, -0.210 ,

respectively). These effect sizes are smaller relative to Model 3. Gender was a significant predictor of math scores ($\beta = -0.100$), but not Spring ODRs or ELA scores. That is, controlling for the screeners accounted for the gender differences in ODRs and ELA scores and created new differences such that females now had lower math achievement scores.

4 | Discussion

Existing EWS data structures, designed to identify students at risk of dropout, often rely on attendance, behavior, and course performance data. While useful, recent studies and policies suggest that these indicators could be complemented with a student self-report behavior screener, particularly as part of an

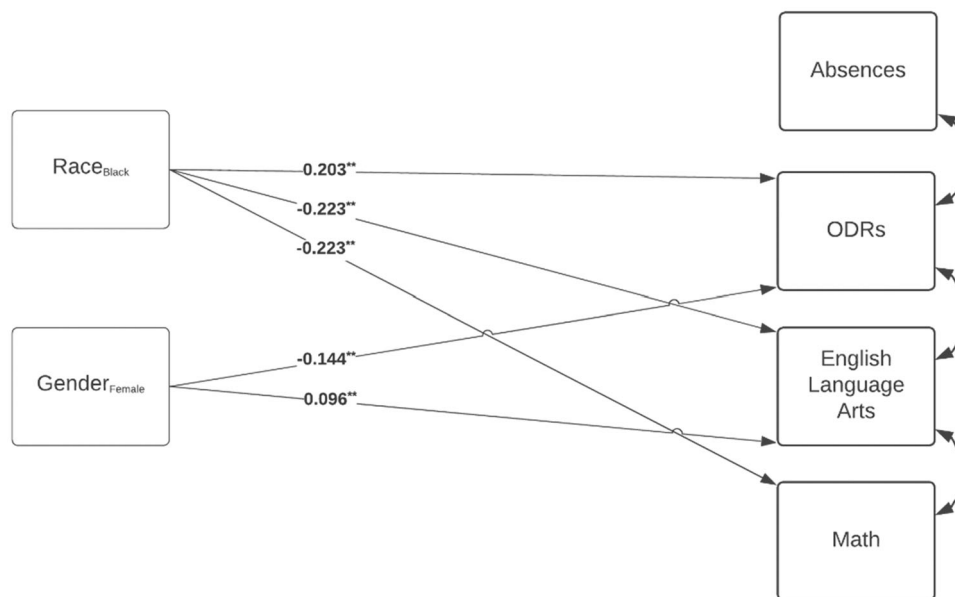


FIGURE 7 | Model 3 predicting attendance, behaviors, and ELA and math performance from the race and gender.

early universal screening process (e.g., Riglin et al. 2014). This will help schools add preventative measures to identify and support at-risk students. However, existing studies have not directly tested how specific behavioral and emotional risk indicators (i.e., behavior screener subscales) are associated with the ABCs, a crucial step for understanding the utility of integrating student self-report screening into an EWS.

Our study focused on student self-report screening data collected within middle schools. We explored the diverging associations between the subscales of a self-report screener, the SDQ, and the ABC metrics used in traditional EWSs. We also tested whether these subscales predict these end-of-year outcomes over and above another early possible EWS metric, Fall ODRs. Finally, we tested whether the subscales account for some of the variance in ABC attributed to race and gender, thus potentially accounting for some of the race and gender bias that could be associated with the ABCs.

Our findings indeed show that different subscales have different relationships with ABC metrics. The conduct problems subscale on the SDQ had the strongest association with the ABC data and predicted poorer student outcomes across all metrics. These findings align with prior studies suggesting that students with behavioral concerns are more likely to display behaviors that result in disciplinary action (Kearney 2002; Lane et al. 2013).

However, hyperactivity-inattention was predictive of higher ELA and math scores and did not predict absences or ODRs. Testing for a possible suppression effect revealed that when not accounting for conduct problems, hyperactivity-inattention predicted more absences and behavior problems but was still predictive of higher achievement. This surprising result is contrary to findings on the negative impact of externalizing symptoms on achievement and other ABC outcomes (e.g., Masten et al. 2005; Zimmermann et al. 2013), and specifically the role of hyperactivity-inattentiveness in this relationship (Galéra et al. 2009). However, more recent works suggest that

evidence on the link between hyperactivity-inattention symptoms and achievement is inconclusive (see a review in Kulkarni et al. 2021) and is especially weak when achievement is measured using standardized testing as opposed to teachers' assessment (e.g., Okano et al. 2020), as in our case. While this does not explain why our findings show a positive (rather than null) relationship between hyperactivity-inattention and achievement, it shows that this relationship is complex and warrants further exploration.

The findings concerning the emotional problems and peer problems subscales were surprising, as well. While others find that internalizing symptoms are associated with fewer negative outcomes across the ABC indicators (e.g., Graybill et al. 2022), our findings show a mixed relationship: emotional problems like stress and anxiety were associated with fewer ODRs and higher ELA scores, but peer problems were only associated with lower ELA scores. This further demonstrates how ABC outcomes may be unlikely to be able to identify students with internalizing issues, further supporting the use of student self-report screeners as a unique contributor to the identification of at-risk students.

We also found that most of these relationships remained significant and relatively strong even after controlling for Fall ODRs. While early ODRs could detect at-risk students early in a less resource-intensive way, they do not capture the full range of emotional and behavioral risks identified by tools like the SDQ. In particular, although the conduct problems scale should measure similar behaviors to those manifested in ODRs, the scale seems to capture slightly different risk factors that predict poorer student outcomes over and above Fall ODRs. That is, our findings support the use of a universal screener for identifying and ideally intervening with students at risk of various behavior and academic problems. Furthermore, behavior screening data are available earlier in the school year, allowing for earlier implementation of prevention and early intervention programs for students identified as at

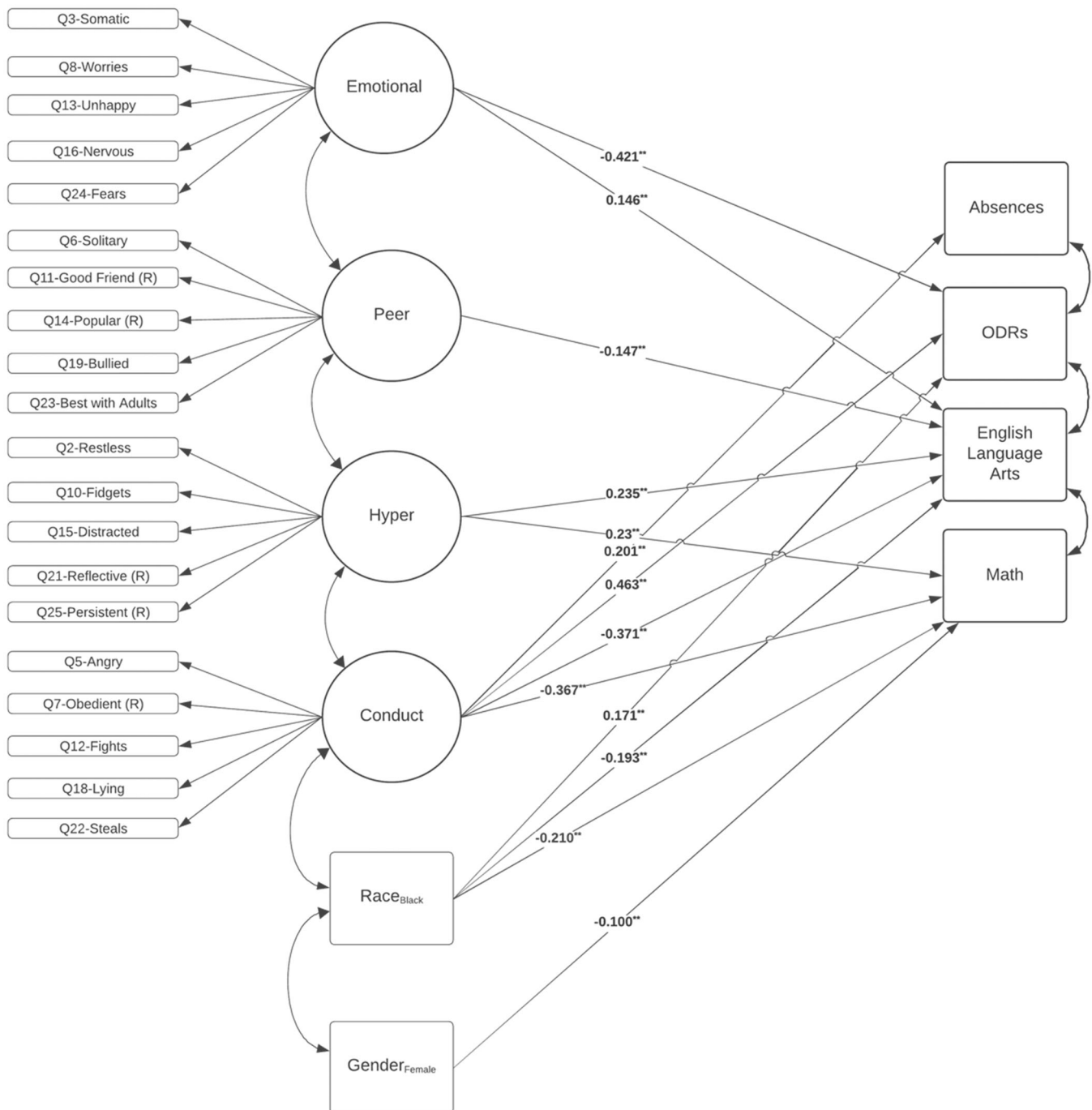


FIGURE 8 | Model 4 predicting attendance, behaviors, and ELA and math performance from race, gender, and the four SDQ subscales. (R) indicates a reverse-coded item.

risk. Therefore, the results of this study point to universal screeners as the better alternative.

Finally, we found that the screener subscales account for some of the variance in ABC outcomes related to race and gender. ABC outcomes associated with ODRs seem consistently biased in favor of White and female students (e.g., Bradshaw et al. 2010; Martinez et al. 2016). This seems to be the case in teacher-reported behavioral screeners, as well (e.g., Kulkarni and Sullivan 2022). Our findings show that student-reported emotional and behavioral health issues can account for the association between ABC outcomes and demographics, again

supporting their use to complement traditional EWS to reduce inequities in schools' prevention of dropout. Universal self-report behavior screeners provide districts with a preventative, proactive and systematically collected data set to inform unbiased decision-making about students (Tara C. Raines et al. 2012).

In addition to the aforementioned strengths of this study, it extends the literature by focusing on a sample of majority Black, low-SES schools, contributing to the findings' generalizability to high-need schools, which may not exhibit the same patterns as other schools. For instance, Young et al. (2021) who focused on

a suburban, mostly White sample did not find any associations between internalizing scores and ABCs.

This study demonstrated that universal behavior screeners like the SDQ provide valuable early identification of students at risk for social, emotional, and behavioral problems, potentially leading to more effective interventions to prevent dropout and later mental health concerns. While it is important to consider the challenges in implementing a universal screening process, including the need for resources and other logistical challenges (Villarreal and Peterson 2024), we found that incorporating these screeners into EWS can help schools proactively address student needs before significant issues develop, aligning with schools' goals to reduce dropout rates and improve student outcomes (Cruden et al. 2016).

Yet, the study was not without limitations. First, we did not test whether the screener contributes to the prediction of dropout beyond traditional ABC outcomes. As this is the central reason for implementing EWSs, such analyses are required before drawing conclusions regarding the utility of universal screening for enhancing EWSs and contributing to early intervention that prevents dropout. Second, while our study's unique design and sample are strengths that can enrich the existing literature (e.g., using a student-report instrument, a sample of mostly Black, low-income, middle school students), the results might not generalize to other contexts. The decision on whether to implement universal screening in particular schools should, therefore, rely on evidence relevant to the schools' characteristics. The sample's uniqueness might also account for our unusual findings concerning hyperactivity-inattentiveness, further supporting the need for future research in high-need schools. Finally, school dropout is a systemic issue influenced by disparities in access to resources within certain student populations. Future research could examine data reflective of those social determinants that predict school completion.

In spite of these limitations, the study's findings have multiple implications for school psychology. The data reveal the complexity of the relationship between different types of behavioral and emotional symptoms and observable ABC data, enhancing our understanding of early mental health issues and their relationships to student outcomes. The results also demonstrate how symptoms and outcomes correlate in different populations and when using different measures, providing guidance for exploring this issue in different contexts. Finally, it adds to the evidence of the potential of student self-report universal screening to complement traditional EWSs within districts' public health tiered system of support models. Schools should, therefore, consider implementing student self-report behavior screening to enhance their dropout prevention programs, especially if the screening can be universal and early.

Acknowledgments

This paper was developed in part under grant number SM083659 from the Substance Abuse and Mental Health Services Administration (SAMHSA), U.S. Department of Health and Human Services (HHS). The views, policies, and opinions expressed are those of the authors and do not necessarily reflect those of SAMHSA or HHS.

Ethics Statement

All procedures have been approved by Rutgers University's IRB (protocol number 000043).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data are confidential and cannot be shared without permission from the school district that provided them.

References

- Archambault, I., M. Janosz, J. Morizot, and L. Pagani. 2009. "Adolescent Behavioral, Affective, and Cognitive Engagement in School: Relationship to Dropout." *Journal of School Health* 79, no. 9: 408–415.
- Assari, S., A. Mardani, M. Maleki, S. Boyce, and M. Bazargan. 2021. "Black-White Achievement Gap: Role of Race, School Urbanity, and Parental Education." *Pediatric Health, Medicine, and Therapeutics* 12: 1–11. <https://doi.org/10.2147/PHMT.S238877>.
- Balfanz, R., and V. Byrnes. 2019. "Early Warning Indicators and Intervention Systems: State of the Field." In *Handbook of Student Engagement Interventions*, edited by J. Fredericks, A. Reschly, and S. Christenson, 45–55. Elsevier Science & Technology.
- Balfanz, R., L. Herzog, and D. J. Mac Iver. 2007. "Preventing Student Disengagement and Keeping Students on the Graduation Path in Urban Middle-Grades Schools: Early Identification and Effective Interventions." *Educational Psychologist* 42, no. 4: 223–235. <https://doi.org/10.1080/00461520701621079>.
- Barger, B., E. Graybill, A. Roach, and K. Lane. 2020. "Differential Item and Test Functioning of the SRSS-IE12 Across Race, Ethnicity, Gender, and Elementary Level." *Assessment for Effective Intervention* 47, no. 2: 79–88. <https://doi.org/10.1177/1534508420976830>.
- Bradshaw, C. P., M. M. Mitchell, L. M. O'Brennan, and P. J. Leaf. 2010. "Multilevel Exploration of Factors Contributing to the Overrepresentation of Black Students in Office Disciplinary Referrals." *Journal of Educational Psychology* 102, no. 2: 508–520. <https://doi.org/10.1037/a0018450>.
- Bruce, M., J. M. Bridgeland, J. H. Fox, and R. Balfanz. 2011. *The Use of Early Warning and Intervention Systems to Build a Grad Nation*. Center for Organization of Schools. <https://eric.ed.gov/?id=ED526421>.
- Butterworth, P., and L. S. Leach. 2018. "Early Onset of Distress Disorders and High-School Dropout: Prospective Evidence From a National Cohort of Australian Adolescents." *American Journal of Epidemiology* 187, no. 6: 1192–1198. <https://doi.org/10.1093/aje/kwx353>.
- Byrne, B. M. 1994. *Structural Equation Modeling With EQS and EQS/Windows: Basic Concepts, Applications, and Programming*. Sage.
- Cavanaugh, B. 2016. "A Preliminary Investigation Examining the Use of Minor Discipline Referral Data to Identify Students at Risk for Behavioral Difficulties: Observations Within Systems of Schoolwide Positive Behavior Interventions and Supports." *Journal of Applied School Psychology* 32, no. 4: 354–366. <https://doi.org/10.1080/15377903.2016.1207739>.
- Chin, M. J., D. M. Quinn, T. K. Dhaliwal, and V. S. Lovison. 2020. "Bias in the Air: A Nationwide Exploration of Teachers' Implicit Racial Attitudes, Aggregate Bias, and Student Outcomes." *Educational Researcher* 49, no. 8: 566–578. <https://doi.org/10.3102/0013189X20937240>.
- Cruden, G., K. Kelleher, S. Kellam, and C. H. Brown. 2016. "Increasing the Delivery of Preventive Health Services in Public Education." Supplement, *American Journal of Preventive Medicine* 51, no. 4 Suppl 2: S158–S167. <https://doi.org/10.1016/j.amepre.2016.07.002>.

- Davis, M. H., M. A. M. Iver, R. W. Balfanz, M. L. Stein, and J. H. Fox. 2019. "Implementation of an Early Warning Indicator and Intervention System." *Preventing School Failure* 63, no. 1: 77–88. <https://doi.org/10.1080/1045988X.2018.1506977>.
- Drummond, T. 1994. *The Student Risk Screening Scale (SRSS)*. Josephine County Mental Health Program.
- Eklund, K., T. L. Renshaw, E. Dowdy, et al. 2009. "Early Identification of Behavioral and Emotional Problems in Youth: Universal Screening Versus Teacher-Referral Identification." *California School Psychologist* 14, no. 1: 89–95. <https://doi.org/10.1007/bf03340954>.
- Freudenberg, N., and J. Ruglis. 2007. "Reframing School Dropout as a Public Health Issue." *Preventing Chronic Disease* 4, no. 4: 107.
- Galéra, C., M. Melchior, J. F. Chastang, M. P. Bouvard, and E. Fombonne. 2009. "Childhood and Adolescent Hyperactivity-Inattention Symptoms and Academic Achievement 8 Years Later: The GAZEL Youth Study." *Psychological Medicine* 39, no. 11: 1895–1906. <https://doi.org/10.1017/S0033291709005510>.
- Goodman, R. 1997. "The Strengths and Difficulties Questionnaire: A Research Note." *Journal of Child Psychology and Psychiatry, and Allied Disciplines* 38, no. 5: 581–586. <https://doi.org/10.1111/j.1469-7610.1997.tb01545.x>.
- Goodman, R. 2001. "Psychometric Properties of the Strengths and Difficulties Questionnaire." *Journal of the American Academy of Child and Adolescent Psychiatry* 40, no. 11: 1337–1345. <https://doi.org/10.1097/00004583-200111000-00015>.
- Graybill, E., A. Roach, and B. Barger. 2020. "Factor Structure of the Self-Report Strength and Difficulties Questionnaire in a Diverse U.S. Sample." *Journal of Psychopathology and Behavioral Assessment* 43, no. 2: 388–398. <https://doi.org/10.1007/s10862-020-09833-4>.
- Graybill, E., A. Salmon, B. Barger, and A. T. Roach. 2022. "Examining the Predictive Utility of the Self-Report Strengths and Difficulties Questionnaire With Middle School Students." *International Journal of Mental Health*: 1–13. <https://doi.org/10.1080/00207411.2022.2038983>.
- Gregory, C., E. C. Graybill, B. Barger, A. T. Roach, and K. Lane. 2020. "Predictive Validity of the Student Risk Screening Scale-Internalizing and Externalizing (SRSS-IE) Scores." *Journal of Emotional and Behavioral Disorders* 29, no. 2: 105–112. <https://doi.org/10.1177/1063426620967283>.
- Heller, S., H. A. Pollack, R. Ander, and J. Ludwig. 2013. *Preventing Youth Violence and Dropout: A Randomized Field Experiment (No. w19014)*. National Bureau of Economic Research.
- Hjorth, C. F., L. Bligrav, L. S. Frandsen, et al. 2016. "Mental Health and School Dropout Across Educational Levels and Genders: A 4.8-year Follow-Up Study." *BMC Public Health* 16: 976. <https://doi.org/10.1186/s12889-016-3622-8>.
- Izumi, J. T., and K. Eklund. 2023. "Universal Screening for Social-Emotional and Behavioral Risk: Differential Item Functioning on the SAEBRs." *School Psychology* 38: 238–246. <https://doi.org/10.1037/spq/0000542>.
- Johfre, S. S., and J. Freese. 2021. "Reconsidering the Reference Category." *Sociological Methodology* 51, no. 2: 253–269. <https://doi.org/10.1177/0081175020982632>.
- Jones, C., E. Graybill, B. Barger, and A. T. Roach. 2020. "Examining the Predictive Validity of Behavior Screeners Across Measures and Respondents." *Psychology in the Schools* 57: 923–936. <https://doi.org/10.1002/pits.22371>.
- Kearney, C. A. 2002. "Identifying the Function of School Refusal Behavior: A Revision of the School Refusal Assessment Scale." *Journal of Psychopathology and Behavioral Assessment* 24, no. 4: 235–245. <https://doi.org/10.1023/A:1020774932043>.
- Kline, R. B. 2015. *Principles and Practice of Structural Equation Modeling* (Fourth Edition. Guilford Publications).
- Kulkarni, T., and A. L. Sullivan. 2022. "Academic Achievement and Relations to Externalizing Behavior: Much Ado About Nothing?" *Journal of School Psychology* 94: 1–14. <https://doi.org/10.1016/j.jsp.2022.07.004>.
- Kulkarni, T., A. L. Sullivan, and J. Kim. 2021. "Externalizing Behavior Problems and Low Academic Achievement: Does a Causal Relation Exist?" *Educational Psychology Review* 33, no. 3: 915–936. <https://doi.org/10.1007/s10648-020-09582-6>.
- Lane, K. L., W. P. Oakes, E. W. Carter, W. E. Lambert, and A. B. Jenkins. 2013. "Initial Evidence for the Reliability and Validity of the Student Risk Screening Scale for Internalizing and Externalizing Behaviors at the Middle School Level." *Assessment for Effective Intervention* 39, no. 1: 24–38. <https://doi.org/10.1177/1534508413489336>.
- Lansford, J. E., K. A. Dodge, G. S. Pettit, and J. E. Bates. 2016. "A Public Health Perspective on School Dropout and Adult Outcomes: A Prospective Study of Risk and Protective Factors From Age 5 to 27." *Journal of Adolescent Health* 58, no. 6: 652–658. <https://doi.org/10.1016/j.jadohealth.2016.01.014>.
- Lawrence, K. C., and T. A. Adebawale. 2023. "Adolescence Dropout Risk Predictors: Family Structure, Mental Health, and Self-Esteem." *Journal of Community Psychology* 51: 120–136. <https://doi.org/10.1002/jcop.22884>.
- Li, C.-H. 2015. "Confirmatory Factor Analysis With Ordinal Data: Comparing Robust Maximum Likelihood and Diagonally Weighted Least Squares." *Behavior Research Methods* 48, no. 3: 936–949. <https://doi.org/10.3758/s13428-015-0619-7>.
- Liu, J., M. S. Hayes, and S. Gershenson. 2024. "JUE Insight: From Referrals to Suspensions: New Evidence on Racial Disparities in Exclusionary Discipline." *Journal of Urban Economics* 141: 103453. <https://doi.org/10.1016/j.jus.2022.103453>.
- Lovelace, M. D., A. L. Reschly, and J. J. Appleton. 2017. "Beyond School Records: The Value of Cognitive and Affective Engagement In Predicting Dropout and On-Time Graduation." *Professional School Counseling* 21, no. 1: 1096–2409. <https://doi.org/10.5330/1096-2409-21.1.70>.
- Margherio, S. M., S. W. Evans, and J. S. Owens. 2019. "Universal Screening In Middle and High Schools: Who Falls through the Cracks." *School Psychology* 34, no. 6: 591–602. <https://doi.org/10.1037/spq000037>.
- Martinez, A., S. D. McMahon, and S. Treger. 2016. "Individual-and School-Level Predictors of Student Office Disciplinary Referrals." *Journal of Emotional and Behavioral Disorders* 24, no. 1: 30–41. <https://doi.org/10.1177/1063426615588289>.
- Masten, A. S., G. I. Roisman, J. D. Long, et al. 2005. "Developmental Cascades: Linking Academic Achievement and Externalizing and Internalizing Symptoms Over 20 Years." *Developmental Psychology* 41, no. 5: 733–746. <https://doi.org/10.1037/0012-1649.41.5.733>.
- McIntosh, K., J. L. Frank, and S. A. Spaulding. 2010. "Establishing Research-Based Trajectories of Office Discipline Referrals for Individual Students." *School Psychology Review* 39, no. 3: 380–394. <https://doi.org/10.1080/02796015.2010.12087759>.
- National Center for Educational Statistics. 2017. *Trends in High School Dropout and Completion Rates in the United States*. Institute of Education Sciences. https://nces.ed.gov/programs/dropout/ind_02.asp#:~:text=2017%20status%20dropout%20rates%20by%20race/ethnicity,the%20rate%20for%20Asian%20youth.
- National Center for Education Statistics. 2023. *Status Dropout Rates. Condition of Education*. U.S. Department of Education, Institute of Education Sciences. <https://nces.ed.gov/programs/coe/indicator/coj>.
- Neild, R. C., R. Balfanz, and L. Herzog. 2007. "An Early Warning System." *Educational Leadership* 65, no. 2: 28–33.
- Okano, L., L. Jeon, A. Crandall, T. Powell, and A. Riley. 2020. "The Cascading Effects of Externalizing Behaviors and Academic

- Achievement Across Developmental Transitions: Implications for Prevention and Intervention." *Prevention Science: The Official Journal of the Society for Prevention Research* 21, no. 2: 211–221. <https://doi.org/10.1007/s11121-019-01055-9>.
- Pinkus, L. 2008. *Using Early-Warning Data to Improve Graduation Rates: Closing Cracks in the Education System*. Alliance for Excellent Education.
- Regional Educational Laboratory Pacific. 2016. *Four Signs Your District is Ready for an Early Warning System*. <https://files.eric.ed.gov/fulltext/ED577836.pdf>.
- Riglin, L., K. V. Petrides, N. Frederickson, and F. Rice. 2014. "The Relationship Between Emotional Problems and Subsequent School Attainment: A Meta-Analysis." *Journal of Adolescence* 37: 335–346. <https://doi.org/10.1016/j.adolescence.2014.02.010>.
- Rodriguez, L. F., and G. Q. Conchas. 2009. "Preventing Truancy and Dropout Among Urban Middle School Youth: Understanding Community-Based Action From the Student's Perspective." *Education and Urban Society* 41, no. 2: 216–247. <https://doi.org/10.1177/0013124508325681>.
- Rosseel, Y. 2012. "Lavaan: An R Package for Structural Equation Modeling." *Journal of Statistical Software* 48, no. 2: 1. <https://doi.org/10.18637/jss.v048.i02>.
- Shin, R. Q., M. A. Kendall, and E. M. Vera. 2012. "Dropout Prevention: A (re)conceptualization Through the Lens of Social Justice." In *The Oxford Handbook of Prevention in Counseling Psychology*. Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780195396423.013.0014>.
- Tara C. Raines, R., D. Bridget V. Dever, K. Randy W. Kamphaus, and R. Andrew T. Roach. 2012. "Universal Screening for Behavioral and Emotional Risk: A Promising Method for Reducing Disproportionate Placement In Special Education." *Journal of Negro Education* 81, no. 3: 283–296.
- U.S. Department of Education. 2016, September. *Issue Brief: Early Warning Systems*. Office of Planning, Evaluation and Policy Development, Policy and Program Studies Service. <https://www2.ed.gov/rschstat/eval/high-school/early-warning-systems-brief.pdf>.
- de Vasconcelos, A. N., L. A. Freires, G. D. L. Loureto, et al. 2023. "Advancing School Dropout Early Warning Systems: The IAFREE Relational Model for Identifying At-Risk Students." *Frontiers in Psychology* 14: 1189283. <https://doi.org/10.3389/fpsyg.2023.1189283>.
- Villarreal, V., and L. S. Peterson. 2024. "Mental Health Screening: Recommendations From an Integrated Literature Review." *Contemporary School Psychology* 29: 250–260. <https://doi.org/10.1007/s40688-024-00501-y>.
- Wentworth, L., and J. Nagaoka. 2020. "Early Warning Indicators in Education: Innovations, Uses, and Optimal Conditions for Effectiveness." *Teachers College Record: The Voice of Scholarship in Education* 122, no. 14: 1–22. <https://doi.org/10.1177/016146812012201401>.
- Young, E. L., S. E. Moulton, and A. Julian. 2021. "Integrating Social-Emotional-Behavioral Screening With Early Warning Indicators in a High School Setting." *Preventing School Failure: Alternative Education for Children and Youth* 65, no. 3: 255–265. <https://doi.org/10.1080/1045988X.2021.1898319>.
- Zimmermann, F., K. Schütte, P. Taskinen, and O. Köller. 2013. "Reciprocal Effects Between Adolescent Externalizing Problems and Measures of Achievement." *Journal of Educational Psychology* 105, no. 3: 747–761. <https://doi.org/10.1037/a0032793>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.